B.Tech 4th Yr. MID Seminar Presentation
on

Development of Vision Based Terrain Recognition System for Quadruped robot

by
ADITYA KUMAR RAI
(Roll No. 122ME0028)

Under the supervision of

Dr. Ravi Kumar Mandava
Asst. Professor, MED
IIITDM Kurnool

Department of Mechanical Engineering
Indian Institute of Information Technology, Design and Manufacturing
Kurnool (A.P.)

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Introduction

How robotics transform itself ?

Before 1970

- Machines that perform repetitive tasks automatically.
- Programmable manipulators in controlled environments.

  ex. Stanford Arm.

2000

- The science and engineering of intelligent machines that can sense, plan, and act.
- Robots move beyond factories into homes, defense.

  ex. Domestic service robot, Space robots.

2010

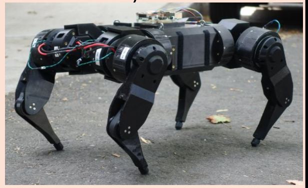
- Design and use of autonomous or semi-autonomous machines that collaborate with humans.
- Human-robot interaction, IoT integration, safer collaboration.

  ex. Medical robot.

2020

- Integration of AI, cloud, and data for adaptive, autonomous machines in dynamic environments.
- Intelligent, data-driven, connected robots.

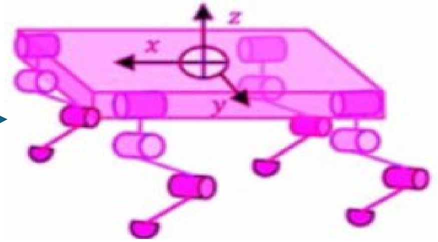
  ex. Quadruped robot, Next-gen medical robots, Aerial robotics.



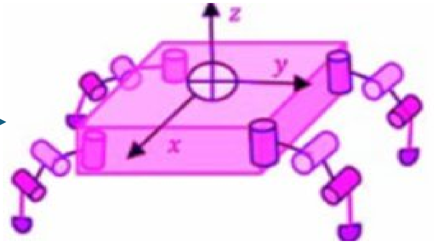
Leg Topologies

Leg Topology

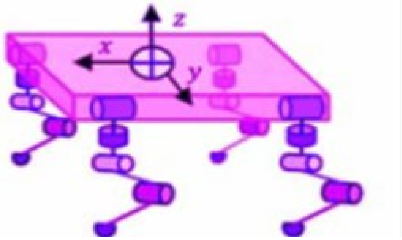
Mammal



Sprawling



Reconfigurable



Quadruped

Joint Configuration

Joint configuration

- Full Symmetric → **Bayes Ant**

- Reversed Symmetric → **HyQ2 Max**

- Half Symmetric → **Jueying**

- Reversed Half symmetric → **GE Quadruped**

Quadruped

What is the need for Quadrupeds?

Criteria	Biped	Quadruped	Hexapod
Number of legs	2	4	6
Inspiration	Human walk	Four-legged animals like dogs and horses etc.	Insects like ants and spiders
Fabrication	S	S	C
Modulation	N	C	S
Stability	C	N	S
Speed	M	H	L
Maneuvering	N	S	C
Cost	M	L	H
Efficiency	L	H	M
Weight	S	N	C
Imitation	S	N	C
Terrain Adaptability	L	M	H

S-Simple, N-Neutral, C-Complex, L-Low, M-Moderate, H-High

Literature Review

S. No.	Author	Title of paper	Working/Finding
1.	Degrave et al.	Terrain Classification for a Quadruped Robot	Terrain classification is found in the combination of tactile sensors and proprioceptive joint angle sensors.
2.	Deng et al.	Vision-based Navigation for a Small-scale Quadruped Robot Pegasus-Mini	Semantic segmentation and navigation in a garden scene are demonstrated to verify the effectiveness of the proposed method.
3.	Biswal et al.	Modeling and Effective Foot Force Distribution for the Legs of a Quadruped Robot	Optimal feet force distribution was calculated through two approaches, and the Euler-Lagrange theory used to derive the joint motion equations, verified by simulation results.
4.	Bakircioglu et al.	Inverse Kinematic Analysis Of A Quadruped Robot	Forward and inverse kinematic analysis of a quadruped robot was investigated.
5.	Atique et al.	Development of an 8DOF quadruped robot and implementation of Inverse Kinematics using Denavit Hartenberg convention	A quadruped robot controlled via an Android app using Bluetooth is developed, featuring six movement modes and obstacle detection with ultrasound sensors.

Literature Review

6.	Haeng Lee et al.	Development of a Quadruped Robot System With Torque-Controllable Modular Actuator Unit	They presented the overall robotic system of AiDIN-VI that has successfully incorporated mandatory abilities.
7.	Fuhai et al.	Omni-directional Quadruped Walking Gaits and Simulation for a Gorilla Robot	In order to realize the quadruped static walking of a gorilla robot, they presented omni-directional quadruped walking gaits.
8.	Xiangrui et al.	A Review of Quadruped Robots and Environment Perception	Several typical and recent robot systems are addressed in their work such as HyQ series, StarlETH, ANYmal, MIT Cheetah and BigDog, etc.

Objectives

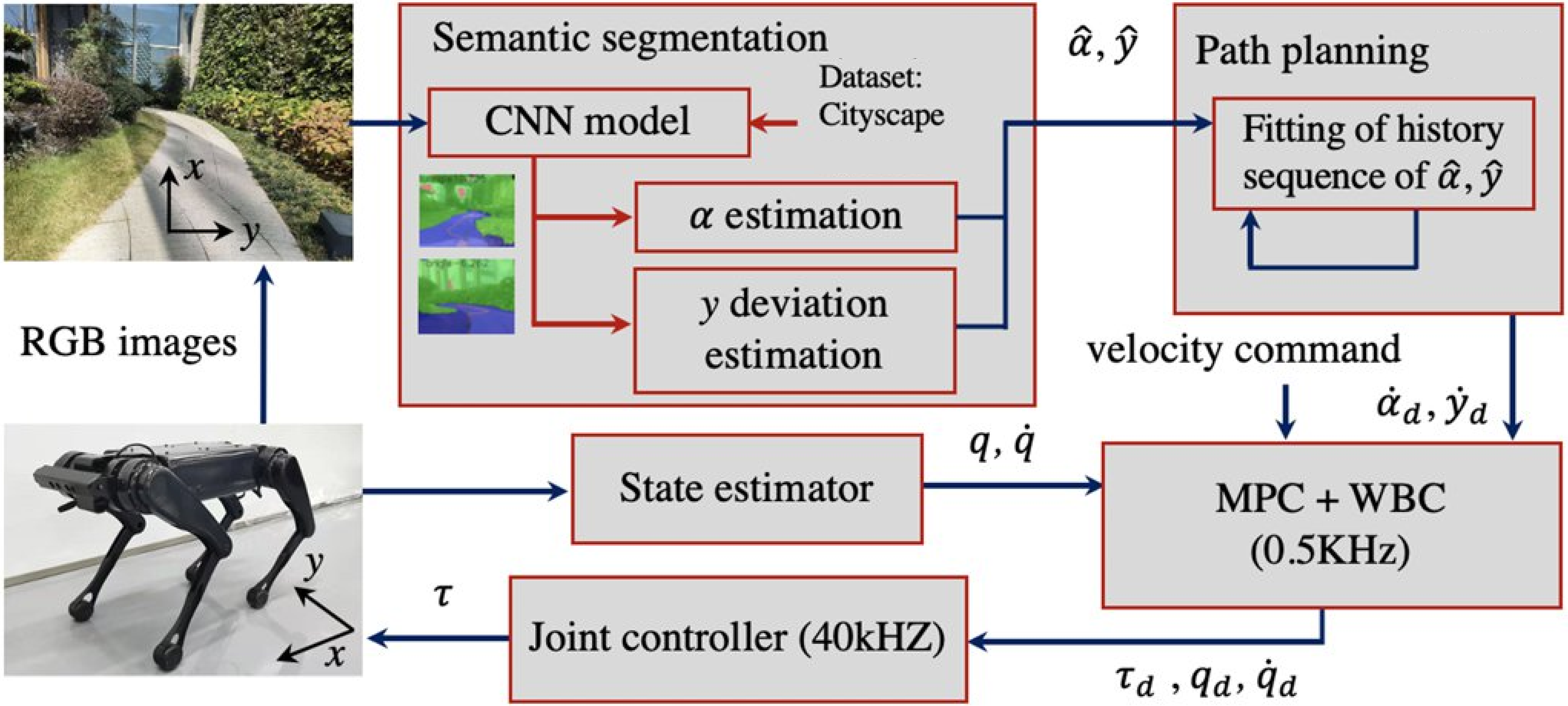
Objective -I :

To integrate Quadruped robot model into a simulation framework and create realistic terrain environment for locomotion testing and data generation .

Objective –II :

Deployment of a YOLO-based CNN model for vision-based terrain classification using Quadruped Robot.

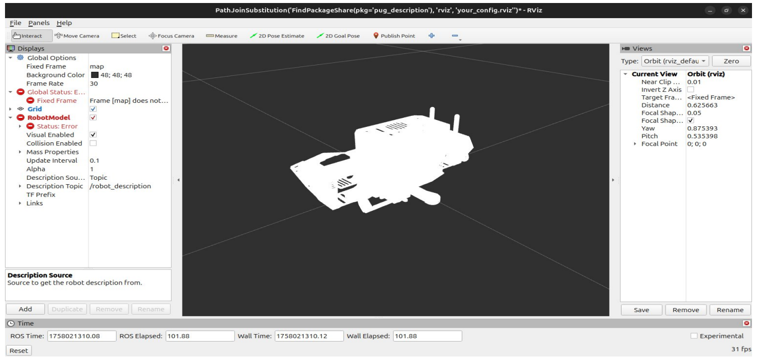
Work Plan and Methodology

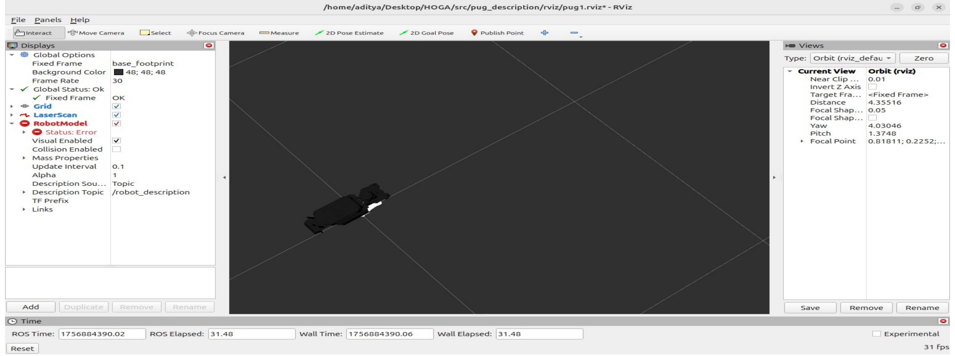


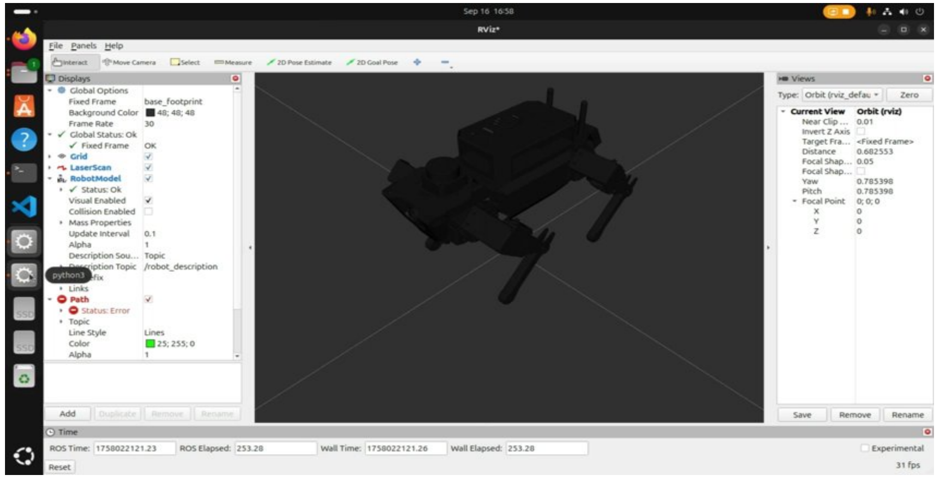
Results and discussion

RViz.

1. The quadruped URDF was successfully launched and rendered in RViz, producing a detailed, interactive 3D visualization of our robot. With the model loaded and TF frames broadcasting, we can inspect link geometry, joint states, and transform relationships in real time — making it easy to verify kinematics, spot configuration issues, and demonstrate the robot's structure to reviewers. This visualization provides a reliable foundation for the next steps: motion testing, sensor integration, and simulation-driven development.



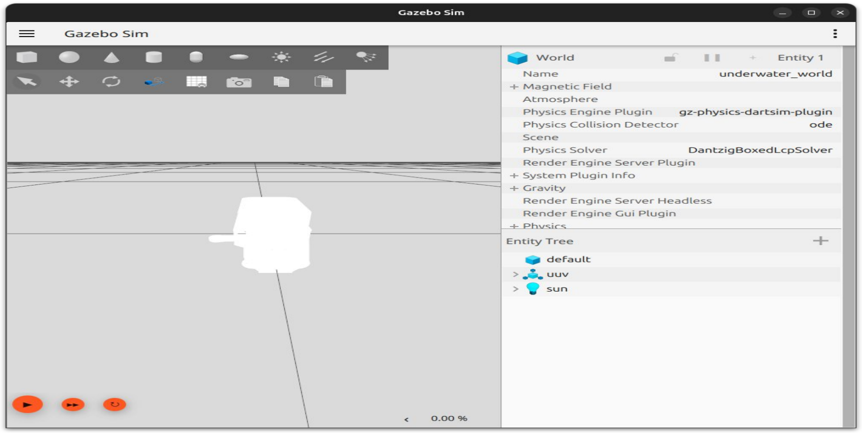


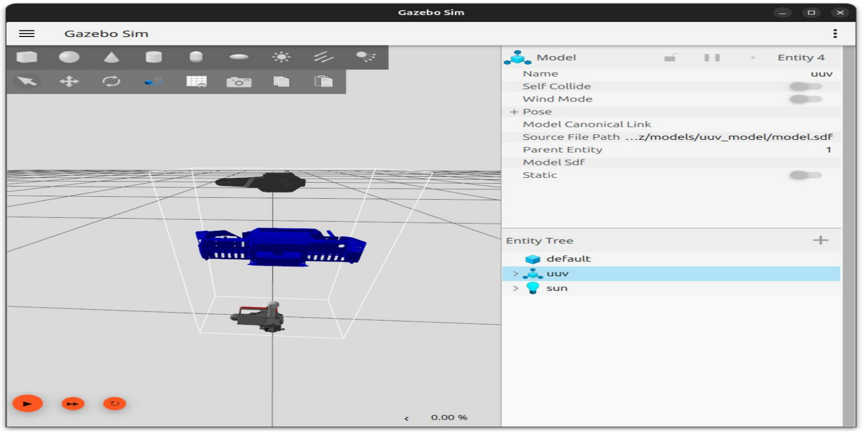


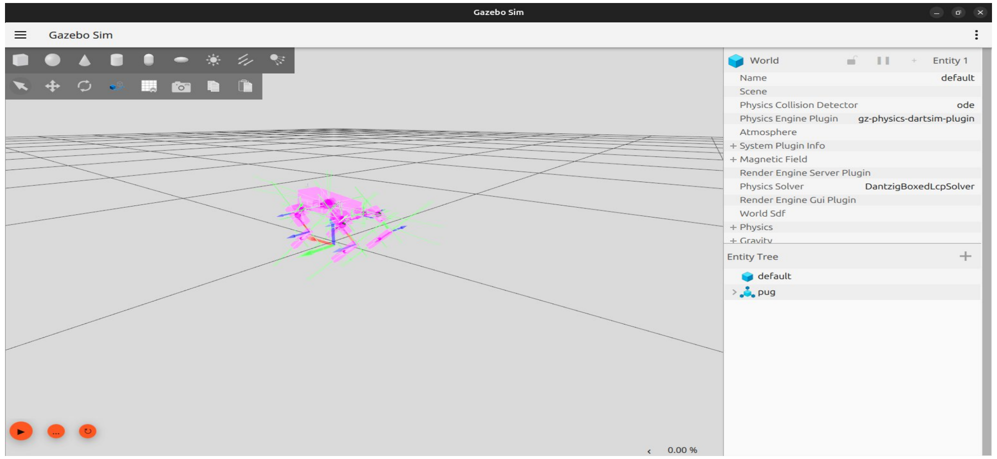
Results and discussion

Gazebo

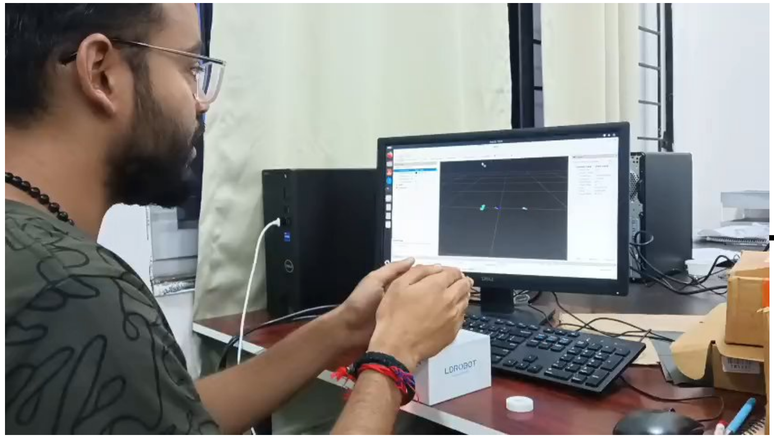
2. Gazebo requires a more detailed URDF or SDF file that includes collision geometries, inertial parameters, and plugin definitions to simulate physics, controllers, and sensors. Gazebo ensures realistic interaction with the environment, making both descriptions essential but tailored to different needs.

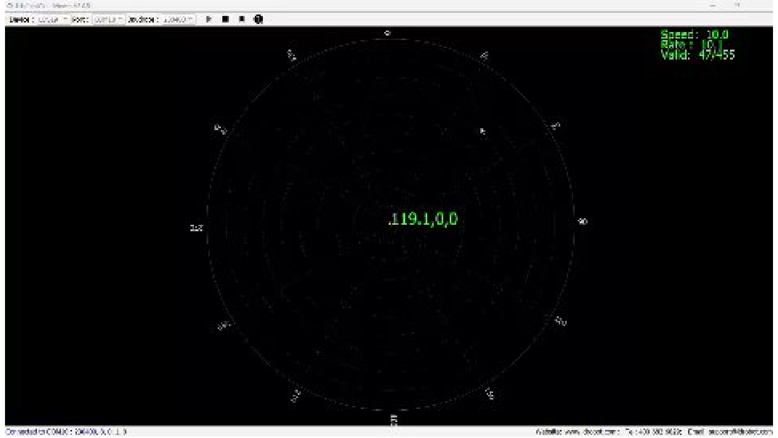






3. In the real time visualization we used Lidar Kit for getting the relative obstacle position from the quadruped robot.





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Thank You